JANE MILLER

Principal Full-Stack AI/ML Engineer

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PROFESSIONAL SUMMARY

Visionary **Full-Stack AI/ML Engineer** with **12+ years** of experience architecting scalable web applications and production-grade machine learning systems. Specializes in bridging the gap between cutting-edge AI research and enterprise-grade web infrastructure. Proven track record of leading cross-functional teams to deliver **20+ production ML models** serving **5M+ daily requests** with **99.97% uptime**. Expert in designing end-to-end MLOps pipelines, building microservices architectures, and implementing real-time AI solutions. Passionate about democratizing AI through robust DevOps practices and creating developer-friendly platforms.

TECHNICAL SKILLS

Frontend

React 18, Next.js 14, TypeScript, Redux Toolkit, Tailwind CSS, Framer Motion, Webpack, Vite

Backend

Python (FastAPI, Django, Flask), Node.js (Express, NestJS), Go (Gin, Fiber), Java (Spring Boot), GraphQL, REST API, gRPC

AI/ML

PyTorch, TensorFlow 2.x, Hugging Face Transformers, LangChain, LlamaIndex, Scikit-learn, XGBoost, LightGBM, PySpark MLlib

MLOps

MLflow, Kubeflow, Airflow, Ray, DVC, Weights & Biases, Evidently AI, Prometheus, Grafana

DevOps & Cloud

Kubernetes, Docker, Helm, Terraform, Ansible, Jenkins, GitLab CI/CD, GitHub Actions, ArgoCD, AWS, Azure, GCP

Data & Databases

Apache Spark, Kafka, Flink, Airflow, dbt, Snowflake, BigQuery, PostgreSQL, MongoDB, Redis, Elasticsearch, Pinecone

PROFESSIONAL EXPERIENCE

TechForge AI

2021 – Present

Senior AI/ML Engineering Lead

San Francisco, CA

- Architected and deployed **15+ production-grade ML models** serving **2M+ daily inference requests** with **p99 latency < 50ms** using PyTorch, FastAPI, and Kubernetes.
- Built an end-to-end **MLOps platform** integrating MLflow, Kubeflow, and Airflow, reducing model deployment time from **weeks to hours** for 30+ data scientists.
- Designed a **multi-cloud hybrid architecture** (AWS + GCP) using Terraform and Kubernetes, achieving **99.99% availability** and saving **$200K/year** in infrastructure costs.
- Led a team of **12 engineers** across frontend, backend, and ML, delivering a **Generative AI document analysis platform** using LangChain and LlamaIndex with 95% accuracy.
- Implemented **real-time model monitoring** with Evidently AI and Prometheus, detecting data drift within **5 minutes** and enabling automatic model retraining.
- Developed a **microservices architecture** with **25+ services** using FastAPI, Go, and gRPC, handling **10K+ concurrent requests** with **Kafka** for event-driven communication.

PyTorch Kubernetes MLflow FastAPI LangChain Terraform Kafka React

CloudSphere Technologies

2017 – 2021

Senior Full-Stack / DevOps Engineer

Austin, TX

- Architected and built a **cloud-native SaaS platform** serving **500+ enterprise clients** with a React frontend, Node.js backend, and PostgreSQL database, scaled to **1M+ active users**.
- Designed a **CI/CD pipeline** using Jenkins, GitHub Actions, and ArgoCD, enabling **50+ deployments per day** with zero-downtime rollbacks.
- Implemented **Infrastructure as Code** with Terraform and Ansible across AWS and Azure, reducing infrastructure provisioning time by **85%**.
- Built a **real-time analytics dashboard** using Elasticsearch, Logstash, and Kibana (ELK stack) processing **10TB+ of logs daily**.
- Containerized all services with Docker and orchestrated using Kubernetes with Helm charts, achieving **99.95% uptime**.
- Introduced **GitOps** practices with ArgoCD, enabling automated rollbacks and declarative infrastructure management.

React Node.js Kubernetes Terraform Jenkins ELK Stack AWS ArgoCD

DataMind AI Labs

2014 – 2017

Machine Learning Engineer

Boston, MA

- Developed **computer vision models** for medical imaging using TensorFlow and PyTorch, achieving **98.5% accuracy** in detecting anomalies from X-ray images.
- Built and deployed **recommendation systems** using collaborative filtering and neural networks, improving user engagement by **45%**.
- Created a **data pipeline** with Apache Spark and Airflow processing **50M+ records daily** for model training and feature engineering.
- Implemented **model versioning and experiment tracking** using DVC and Weights & Biases, enabling reproducible research.
- Deployed models as **REST APIs** using Flask and Docker, serving predictions to internal teams and external clients.

TensorFlow PyTorch Apache Spark Airflow DVC Flask Docker

WebForge Studios

2012 – 2014

Full-Stack Web Developer

New York, NY

- Designed and developed **20+ client websites and web applications** using HTML5, CSS3, JavaScript, PHP, and MySQL.
- Built custom **CMS platforms** and **e-commerce solutions** with WordPress and Shopify integrations.
- Implemented **responsive design** using Bootstrap and custom CSS, ensuring compatibility across all devices.
- Optimized website performance, achieving **Google PageSpeed scores of 95+** through code minification, lazy loading, and CDN integration.
- Managed **hosting and deployment** on AWS and shared hosting environments.

JavaScript PHP MySQL WordPress AWS Bootstrap

KEY PROJECTS

Intelligent Document Processing (IDP) Platform

Built a **Generative AI platform** that extracts, classifies, and summarizes documents using **LLMs (GPT-4, Claude 3)** and **RAG architecture** with Pinecone vector database. The platform processes **100K+ documents/month** with 97% accuracy, reducing manual review time by 85%.

LangChain LlamaIndex Pinecone FastAPI React Kubernetes

Predictive Maintenance System for Industrial IoT

Designed a **real-time anomaly detection system** using **LSTM neural networks** and **XGBoost** to predict equipment failure 48 hours in advance with 94% precision. Deployed on **Kubernetes** with **Apache Kafka** for real-time data streaming, serving **10K+ sensors** across 50+ manufacturing plants.

PyTorch XGBoost Kafka Kubernetes TimescaleDB Grafana

E-Commerce AI Recommendation Engine

Developed a **hybrid recommendation system** combining collaborative filtering, content-based filtering, and neural networks. The system handles **5M+ users** and **1M+ products**, generating personalized recommendations in **< 100ms** with a **35% increase in conversion rate**.

TensorFlow Redis Elasticsearch Node.js React AWS

Multi-Cloud MLOps Platform

Created a **self-service MLOps platform** enabling data scientists to train, validate, and deploy models with **one-click deployment**. Integrated **MLflow** for tracking, **Kubeflow** for orchestration, **Airflow** for scheduling, and **Prometheus + Grafana** for monitoring. Reduced model deployment time by 90%.

MLflow Kubeflow Airflow Terraform AWS GCP

EDUCATION

M.S. in Computer Science - Stanford University 2012 – 2014

B.S. in Computer Engineering - University of California, Berkeley 2008 – 2012

CERTIFICATIONS

- 🎓 AWS Certified Solutions Architect – Professional
- 🎓 Google Professional Machine Learning Engineer
- 🎓 Kubernetes Certified Administrator (CKA)
- 🎓 HashiCorp Terraform Certified
- 🎓 Microsoft Azure AI Engineer Associate
- 🎓 Professional Scrum Master (PSM I)

PUBLICATIONS & SPEAKING

- **"MLOps at Scale: Building Resilient AI Infrastructure"** – O'Reilly AI Conference, 2023
- **"Bridging the Gap: AI Research to Production"** – PyCon US, 2022
- **"Productionizing Large Language Models with RAG"** – AWS re:Invent, 2023
- **"Kubernetes for ML Workloads"** – KubeCon + CloudNativeCon, 2022
- **"Real-Time Anomaly Detection using Deep Learning"** – IEEE International Conference, 2021

AWARDS & RECOGNITION

- 🏆 **Innovation Award** – TechForge AI (2023) for building the IDP platform
- 🏆 **Best Paper** – IEEE Big Data Conference (2021)
- 🏆 **Engineering Excellence Award** – CloudSphere Technologies (2020)
- 🏆 **Google Developer Expert** – Cloud & ML (2019–Present)

OPEN SOURCE CONTRIBUTIONS

- **LangChain** – Contributed to document loaders and vector store integrations
- **FastAPI** – Added support for background tasks and WebSocket improvements
- **MLflow** – Enhanced model registry and deployment APIs
- **Ray** – Contributed to distributed training optimizations
- Creator of **PyMLOps** – Open-source MLOps toolkit with 2.5K+ stars on GitHub